Educational Philosophy

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The end-of-day study hall I proctored was coming to a close when a student insisted on telling me a joke: "What do Latin and mathematics have in common? They're both dead languages." The intended mathematical jab amused me, but before I could respond my mind raced back to when I was in college. I vividly recalled my Latin professor passionately telling us that Latin was not a dead language but a living, breathing organism and that the purpose of any language is to facilitate communication between its users. As a class, we learned to read, write, and speak Latin over the course of the year. I refocused my attention on the student in front of me who was enrolled in both Latin and math classes at the time. I could tell he was trying to elicit laughter from myself and others, which he accomplished, but I sensed a tinge of seriousness in his voice that caused me to pause and reflect. What would it mean for a student to learn how to read, write, and speak mathematics? I contemplated how mathematics might come alive for a student. What I discovered has shaped how I approach mathematics outside and inside the classroom.

The greatest problem I face each year with any new group of students is getting them to disarm themselves and engage each other in meaningful mathematical discussion. I have found mathematical magic tricks on the first day of class to be an unusually effective solution. If chosen effectively, a magic trick can illustrate the need for and spark the development of a wide variety of important attributes any healthy math class should have. Students practice critical thinking by attempting to determine what is relevant or irrelevant to the magic trick under consideration, and they foster a sense of community by working together in small groups. My role in this process often manifests itself in the form of rotating between groups to ensure that all members are engaged and to help them formulate questions that probe their reasoning and nudge them in the right direction. Due next class is a written analysis of one of the tricks by each student.

The first day of my class sets the stage for all future days and serves as a window into my teaching philosophy as a whole. My primary goal is to help students realize that mathematics, like language, is a *social* enterprise and fundamentally a tool for *human* understanding and communication. In daily practice, this often translates to providing mini-lectures at the beginning of class, where I strive to evoke questions that can serve as the impetus for providing skills and techniques that will aid students in cobbling together their own explanations and proofs. Students typically begin exploring questions in small groups once a mini-lecture has concluded. In my experience, investigative exploration is where students chiefly learn to acquire and practice the language of mathematics. But how can an educator effectively observe student progress throughout such explorations? Bill Thurston said it best: "Mathematics is a transforming journey, and progress in it can be better measured by changes in how we think than by the external truths we discover."

While a student's grade in my class will largely be determined by homework, quizzes, tests, and a writing portfolio when possible, I have found that the most effective and meaningful gauge for student progress comes from observing a student's behavior in the classroom. Are they engaged? What kinds of questions are they asking? Are they progressively thinking in a more sophisticated fashion? To encourage maximal engagement in class, I develop problem sets (with solutions) that are completely self-contained in the hope of eradicating the far too common practice of obsessive notetaking without actual listening. I do not want there to be a substitute for the magic that can transpire in a community that is learning to speak the same language—no document can provide the gestures, sound effects, and body language that characterize human interaction. The classroom simply serves as the venue in which the magic can take place.

The greatest compliment I received by a student was an anonymous response to a feedback question concerning how my class had changed how they thought about math or themselves as a math student (reproduced verbatim): "This class did change my idea about math to some extent. After taking the class, I found maths is not only a subject but a kind of art in some cases. It does have charm, but I have to strive hard to find it. As a math student, I've become more careful, starting paying more attention to details." This student came to experience mathematics as a transformative journey that enhanced his or her way of thinking and understanding. I hope to help others do the same.